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- Python
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A ():

A


```

//| panel: input
viewof length = Inputs.range([0.1, 2.0], {value: 1.0, label: "    (m)", step: 0.1})
viewof initialAngle = Inputs.range([-Math.PI/2, Math.PI/2], {value: Math.PI/4, label: "    (r"
viewof gravity = Inputs.range([1.0, 20.0], {value: 9.8, label: "    (m/s2)", step: 0.1})

//| echo: false
//| fig-cap: "    "

//    60fps
import { now } from "@observablehq/stdlib"

//
state = {
  const t = now / 1000; //
  //    =  $\sqrt{g/l}$ 
  const omega = Math.sqrt(gravity / length);
  //    (t) = 0 * cos(t)
  const angle = initialAngle * Math.cos(omega * t);

  return {
    time: t,
    angle: angle,
    x: length * Math.sin(angle),
    y: length * Math.cos(angle)
  };
}

```

```
//
width = 400
height = 300
cx = width / 2
cy = 50
scale = 100 // 1m = 100px

// SVG
html`<svg width="${width}" height="${height}" style="background-color: #f8f9fa; border: 1px solid #f8f9fa; border-radius: 10px;">
  <!-- -->
  <circle cx="${cx}" cy="${cy}" r="5" fill="#333" />

  <!-- -->
  <line x1="${cx}" y1="${cy}"
        x2="${cx + state.x * scale}" y2="${cy + state.y * scale}"
        stroke="#666" stroke-width="2" />

  <!-- -->
  <circle cx="${cx + state.x * scale}" cy="${cy + state.y * scale}"
          r="15" fill="#e63946" stroke="#333" stroke-width="2" />
</svg>`
```

l m

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \sin \theta$$

$$\theta \ll 1 \quad \sin \theta \approx \theta$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \theta$$

$$\theta(t) = \theta_0 \cos \left(\sqrt{\frac{g}{l}} t + \phi \right)$$

$$\begin{array}{lll}
 r = ae^{k\theta} \ (\theta > 0) & \dot{\theta} = \omega = \text{const} & v_r, v_\theta, a_r, a_\theta \\
 a > 0 & a_r = k &
 \end{array}$$

1.

$$(r, \theta)$$

$$\begin{array}{l}
 \bullet \quad : v_r = \dot{r}, \quad v_\theta = r\dot{\theta} \\
 \bullet \quad : a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}
 \end{array}$$

$$\begin{array}{l}
 \bullet \quad r = ae^{k\theta} \\
 \bullet \quad \dot{\theta} = \omega \quad (\quad) \\
 \bullet \quad \ddot{\theta} = 0
 \end{array}$$

$$r \qquad \dot{\theta} = \omega$$

$$\dot{r} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = (ake^{k\theta}) \cdot \omega = ak\omega e^{k\theta} = k\omega r$$

$$\ddot{r}$$

$$\ddot{r} = \frac{d}{dt}(k\omega r) = k\omega \dot{r} = k\omega(ak\omega e^{k\theta}) = ak^2\omega^2 e^{k\theta}$$

:

$$\begin{aligned}v_r &= \dot{r} = ak\omega e^{k\theta} \\v_\theta &= r\dot{\theta} = (ae^{k\theta})\omega = a\omega e^{k\theta}\end{aligned}$$

:

$$\begin{aligned}a_r &= \ddot{r} - r\dot{\theta}^2 = ak^2\omega^2 e^{k\theta} - (ae^{k\theta})\omega^2 = a\omega^2(k^2 - 1)e^{k\theta} \\a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = (ae^{k\theta})(0) + 2(ak\omega e^{k\theta})\omega = 2ak\omega^2 e^{k\theta}\end{aligned}$$

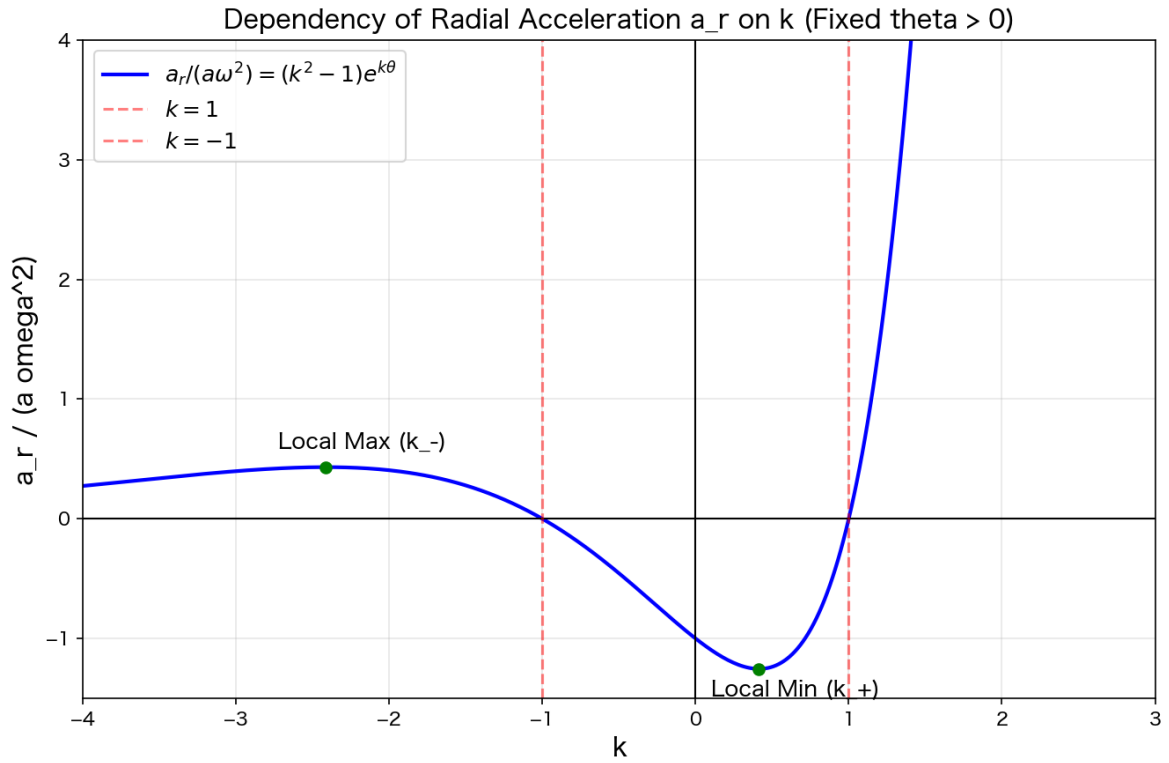
2. $a_r = k$

$$\begin{aligned}a_r(k) &= a\omega^2(k^2 - 1)e^{k\theta} \quad a > 0 \quad \theta > 0 \quad k \quad a\omega^2 > 0 \\f(k) &= (k^2 - 1)e^{k\theta}\end{aligned}$$

$$\begin{aligned}f'(k) &= 2ke^{k\theta} + \theta(k^2 - 1)e^{k\theta} = (\theta k^2 + 2k - \theta)e^{k\theta} \\f'(k) = 0 \quad k = 2 \quad \theta k^2 + 2k - \theta &= 0\end{aligned}$$

$$k_{\pm} = \frac{-1 \pm \sqrt{1 + \theta^2}}{\theta}$$

$$\theta > 0 \quad k_+ > 0 \quad k_- < -1$$



1: a_r k

1. $a_r : e^{k\theta} a_r (k^2 - 1)$

- $|k| < 1 : a_r < 0$
- $|k| > 1 : a_r > 0$
- $k = \pm 1 : a_r = 0$

2. $k \rightarrow \infty : e^{k\theta}$

3. $k \rightarrow -\infty : (k^2 - 1) e^{k\theta} \rightarrow 0 \quad a_r \rightarrow 0$

4. $k < -1 \quad 0 < k < 1$